CHE 201 Spring 2020 HW 7

Roxanna Mendoza

TOTAL POINTS

81 / 100

QUESTION 1

1 1 **12 / 15**

- **- 2 pts** Wrong schematic of the system
- **- 2 pts** Wrong engineering model
- **- 4 pts** wrong mass rate and energy rate balance
- **- 4 pts** Wrong exit temperature
- ✓ **- 3 pts Wrong P-v diagram**
- **- 0 pts** Correct

QUESTION 2

2 2 **15 / 15**

- ✓ **- 0 pts Correct**
- **- 2 pts** Wrong schematic of the system
- **- 2 pts** Wrong engineering model
- **- 4 pts** Wrong mass rate and energy rate balance
- **- 2 pts** Wrong mass flow rate of the air
- **- 2 pts** Wrong exit area
- **- 3 pts** Wrong T-v diagram

QUESTION 3

3 3 **12 / 15**

- **- 0 pts** Correct
- **- 2 pts** Wrong schematic of the system
- **- 2 pts** Wrong engineering model
- **- 4 pts** Wrong mass rate and energy rate balance
- **- 4 pts** Wrong power developed by the turbine
- ✓ **- 3 pts Wrong diagram P-V**
- **- 1 pts** Incomplete engineering model

QUESTION 4

4 4 **12 / 15**

- **- 0 pts** Correct
- **- 2 pts** Wrong schematic of the system
- **- 2 pts** Wrong engineering model
- **- 1 pts** Incompelet engineering model
- **- 4 pts** Wrong mass rate and energy rate balance
- **- 4 pts** Wrong heat transfer rate
- ✓ **- 3 pts Wrong T-v diagram**

QUESTION 5

5 5 **14 / 20**

- **- 0 pts** Correct
- **- 4 pts** Wrong schematic of the system
- **- 4 pts** Wrong engineering model
- **- 6 pts** Wrong mass rate and energy rate balance
- ✓ **- 6 pts Wrong the power required by the pump**
- **- 1 pts** W<0 work done on system

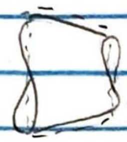
QUESTION 6

6 6 **16 / 20**

- **- 0 pts** Correct
- **- 3 pts** Wrong schematic of the system
- **- 3 pts** Wrong engineering model
- **- 5 pts** Wrong mass rate and energy rate balance
- **- 5 pts** Wrong quality
- ✓ **- 4 pts Wrong T-v diagram**
- **- 2 pts** Incompelet engineering model

CHE 201 Homework #7 Roxanna Mendoza

#1. Nozzle



Engineering Model
1. open system
2. steady state
3. $\Delta PE = 0$
4. well insulated $\dot{Q} = 0$

Mass rate balance

$$\dot{m}_i = \dot{m}_e$$

Energy rate balance

$$0 = \cancel{\dot{Q}_{cv}}^{0} - \cancel{\dot{W}_{cv}}^{0} + \dot{m}\left[(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2}\right) + g\cancel{(z_1 - z_2)}^{0}\right]$$

$$0 = (h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2}\right)$$

knowns:

$P_1 = 200 \text{ lbf/in}^2$ $P_2 = 20 \text{ lbf/in}^2$

$T_1 = 220\,°F$ $T_2 = ?$

$V_1 = 120 \text{ ft/s}$ $V_2 = 1500 \text{ ft/s}$

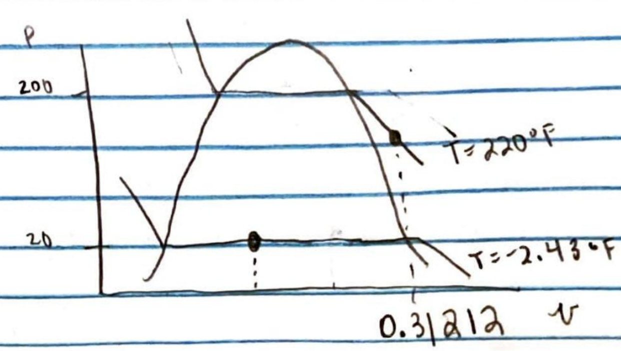
$h_1 = 144.19 \text{ Btu/lbm}$ ← from Table A-11

$$h_2 = h_1 + \left(\frac{V_1^2 - V_2^2}{2}\right)$$

$$h_2 = (144.19) + (0.5(120)^2 - (1500)^2)\left|\frac{1 \text{ Btu}}{778 \text{ ft}\cdot\text{lbf}}\right|\left|\frac{1 \text{ lbf}}{32.2 \text{ lbm}\cdot\text{ft/s}^2}\right|$$

$h_2 = 144.19 - 44.6199 = 99.57 \text{ Btu/lbm}$

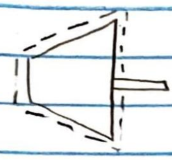
$T_2 = -2.43\,°F$



1 1 **12 / 15**

- **- 2 pts** Wrong schematic of the system
- **- 2 pts** Wrong engineering model
- **- 4 pts** wrong mass rate and energy rate balance
- **- 4 pts** Wrong exit temperature
- ✓ **- 3 pts** **Wrong P-v diagram**
- **- 0 pts** Correct

#2.



Engineering Model

1. open system
2. Steady State
3. $\Delta PE = 0$
4. $Q = 0$
5. Air is an ideal gas

Mass rate Balance:

$$\dot{m}_i = \dot{m}_e$$

Energy rate Balance:

$$0 = \cancel{\dot{Q}_{cv}} - \dot{W}_{cv} + \dot{m}\left[(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2}\right) + \cancel{g(z_1 - z_2)}\right]$$

$$\dot{W}_{cv} = \dot{m}(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2}\right)$$

Knowns:

$P_1 = 96$ bar $\quad P_2 = 1$ bar

$T_1 = 960$ K $\quad T_2 = 450$ K

$V_1 = 0$ m/s $\quad V_2 = 90$ m/s

$P = 2500$ kW

$h_1 = 1000.55$ (kJ/kg) ← from Table A-22

$h_2 = 451.80$ kJ/kg

$$2500 = \dot{m}(1000.55 - 451.80) + \left(\frac{0^2 - 90^2}{2 \cdot 1000}\right)$$

$$\dot{m} = 4.5484 \text{ kg/sec}$$

$$\rho = \frac{P}{RT} \qquad \rho = \frac{(1 \text{ bar})(10^2 \text{ N/m}^2)}{(0.287 \text{ kJ/kg}\cdot\text{K})(450 \text{ K})}$$

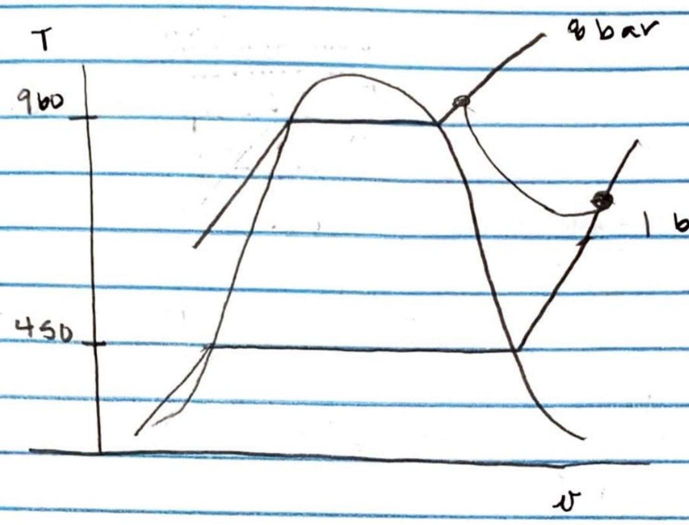
$$\rho = 0.77 \text{ kg/m}^3$$

$$\dot{m} = A \cdot V \cdot \rho$$

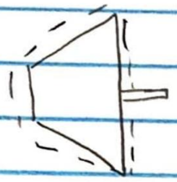
$$4.5484 \frac{\text{kg}}{\text{sec}} = A \cdot (90 \text{ m/s}) \cdot (0.77 \text{ kg/m}^3)$$

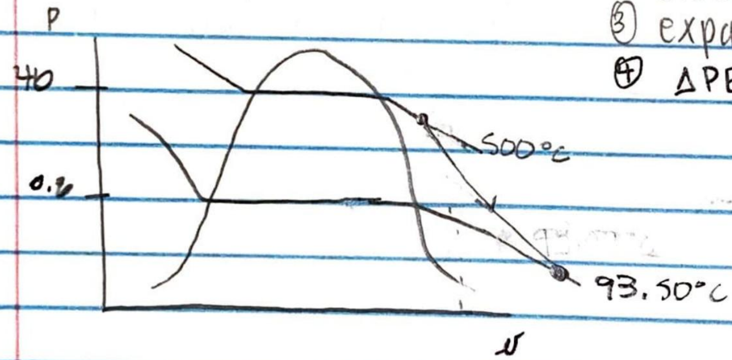
$$A = 0.0656 \text{ m}^2$$

Part E:



#3.





Engineering Model:

① open system
② steady state
③ expands adiabatically
④ $\Delta PE = 0$

knowns:

$T_1 = 500°C$ $P_2 = 0.8\text{ bar}$

$P_1 = 40\text{ bar}$ $V_2 = 150\text{ m/s}$

$V_1 = 200\text{ m/s}$ $A \cdot V = 9.48\text{ m}^3/\text{s}$

mass rate:

$\dot{m}_i = \dot{m}_e$

Energy Rate:

$\dot{W}_{cv} = \dot{m}\left[(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2}\right)\right]$

$h_1 = 3445.3\text{ kJ/kg}$ $v_1 = 0.08643\text{ m}^3/\text{kg}$

$h_2 = 2665.8\text{ kJ/kg}$ $v_2 = 2.087\text{ m}^3/\text{kg}$

$\dot{m} = \frac{9.48}{2.087} = 4.542\ \frac{\text{kg}}{\text{s}}$

$\dot{W}_{cv} = (4.542)(3445.3 - 2665.8) + \left(\frac{200^2 - 150^2}{2}\right)\left(\frac{1}{10^3}\right)$

$\dot{W}_{cv} = 3590.23\text{ kW}$

2 2 **15 / 15**

✓ **- 0 pts** Correct

- 2 pts Wrong schematic of the system

- 2 pts Wrong engineering model

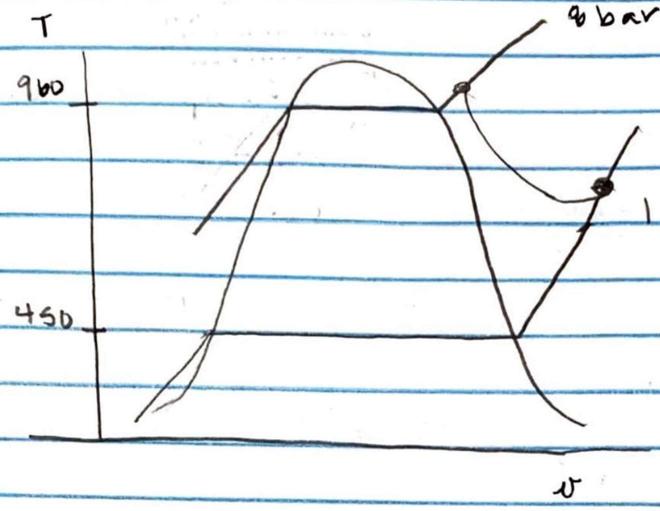
- 4 pts Wrong mass rate and energy rate balance

- 2 pts Wrong mass flow rate of the air

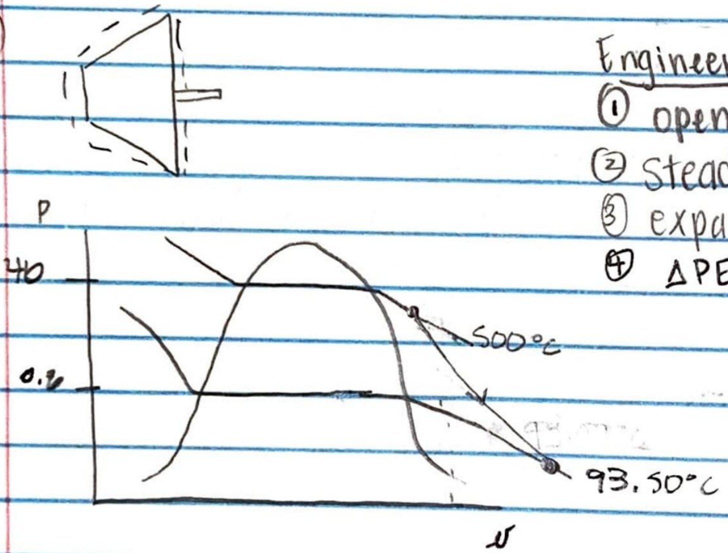
- 2 pts Wrong exit area

- 3 pts Wrong T-v diagram

Part E:



#3.



<u>Engineering Model:</u>

1. open system
2. steady state
3. expands adiabatically
4. $\Delta PE = 0$

knowns:

$T_1 = 500°c$ $\quad P_2 = 0.8 \text{ bar}$

$P_1 = 40 \text{ bar}$ $\quad V_2 = 150 \text{ m/s}$

$V_1 = 200 \text{ m/s}$ $\quad A \cdot V = 9.48 \text{ m}^3/\text{s}$

Mass rate:

$\dot{m}_i = \dot{m}_e$

Energy Rate:

$\dot{W}_{cv} = \dot{m}\left[(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2}\right)\right]$

$h_1 = 3445.3 \text{ kJ/kg}$ $\quad v_1 = 0.08643 \text{ m}^3/\text{kg}$

$h_2 = 2665.8 \text{ kJ/kg}$ $\quad v_2 = 2.087 \text{ m}^3/\text{kg}$

$\dot{m} = \frac{9.48}{2.087} = 4.542 \frac{\text{kg}}{\text{s}}$

$\dot{W}_{cv} = (4.542)(3445.3 - 2665.8) + \left(\frac{200^2 - 150^2}{2}\right)\left(\frac{1}{10^3}\right)$

$\dot{W}_{cv} = 3580.23 \text{ kW}$

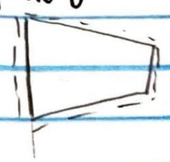
3 3 **12 / 15**

- **- 0 pts** Correct
- **- 2 pts** Wrong schematic of the system
- **- 2 pts** Wrong engineering model
- **- 4 pts** Wrong mass rate and energy rate balance
- **- 4 pts** Wrong power developed by the turbine
- ✓ **- 3 pts** **Wrong diagram P-V**
- **- 1 pts** Incomplete engineering model

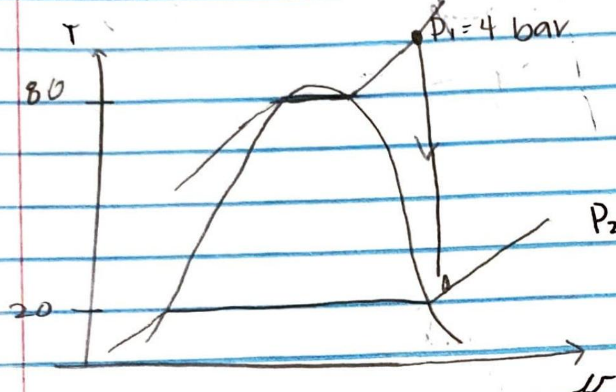
$A.\ \dot{V} = 4\ m^3/min$

$P_1 = 4$ bar
$T_1 = 20°C$

#4.



Engineering Model
① open system
② steady state
③ $\Delta KE = \Delta PE = 0$



$v_1 = 0.05397\ m^3/kg$

$v_2 = 0.02051\ m^3/kg$

Mass rate:
$\dot{m}_i = \dot{m}_e$

energy rate:
$\dot{W}_{cv} = \dot{Q}_{cv} + \dot{m}(h_1 - h_2)$

From Table A-12
$h_1 = 262.96$ KJ/kg
$h_2 = 310.24$ KJ/kg

$W = -60$ KJ per kg

$$\dot{m} = \frac{4\ m^3/min \div 60}{0.05397\ m^3/kg} = 1.235\ \frac{kg}{sec}$$

$\dot{Q}_{cv} = \dot{m}(\dot{W}_{cv} + (h_2 - h_1))$

$\dot{Q}_{cv} = (1.235) * (60 + (310.24 - 262.96))$

$\dot{Q}_{cv} = -15.71$ KW

4 4 12 / 15

- **- 0 pts** Correct
- **- 2 pts** Wrong schematic of the system
- **- 2 pts** Wrong engineering model
- **- 1 pts** Incompelet engineering model
- **- 4 pts** Wrong mass rate and energy rate balance
- **- 4 pts** Wrong heat transfer rate
- ✓ **- 3 pts** **Wrong T-v diagram**

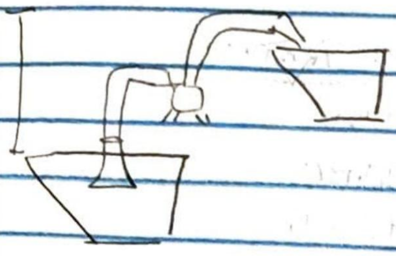
$T_1 = 15°C$
$P = 1$ bar

$P_2 = 3$ bar

$\dot{m} = 1.5 \frac{kg}{s}$ $T_2 = 15°C$

15 m

#5.



Engineering Model
1. open system
2. steady state
3. Temperature is constant
4. $\Delta KE = 0$
5. $Q = 0$

Mass rate:

$$\dot{m}_i = \dot{m}_e$$

energy rate:

$$\dot{W}_{cv} = \dot{m}[(h_1 - h_2) + g(z_1 - z_2)]$$

$$h_1 - h_2 = [h_f(T_1) + v_f(T_1)[p_1 - p_{sat}(T_1)] - h_f(T_2 + v_f(T_2)[p_2 - p_{sat}(T_2)])$$

$$h_1 - h_2 = v_f(T)(p_1 - p_2)$$

$$v_f(T) = 1.0009 \times 10^3 \ m^3/kg$$

$$h_1 - h_2 = (1.0009 \times 10^3)(1-3)$$

$$h_1 - h_2 = -2001.8$$

$$\dot{W}_{cv} = (1.5 \ kg/s)[(-2001.8 \ kJ/kg) + (9.81(15))]$$

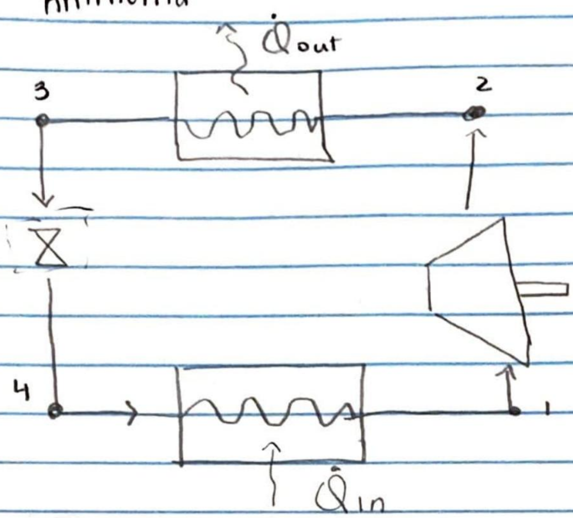
$$\dot{W}_{cv} = -27901.98 \ kW$$

5 5 **14 / 20**

- **- 0 pts** Correct
- **- 4 pts** Wrong schematic of the system
- **- 4 pts** Wrong engineering model
- **- 6 pts** Wrong mass rate and energy rate balance
- ✓ **- 6 pts** **Wrong the power required by the pump**
- **- 1 pts** W<0 work done on system

Ammonia

#6.



Engineering Model

1. open system
2. undergoes throttling process
3. steady state

$P = 10$ bar $P_2 = 1$ bar

$T_1 = 24°C$

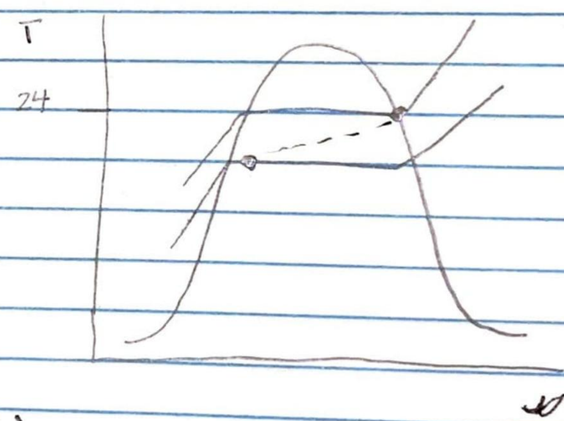
Mass rate Energy rate

$\dot{m}_i = \dot{m}_e$ $h_2 = h_1$

$P_1 = 10$ bar $T_1 = 24°C$

$h_1 = 313.2$ kJ/kg

$h_1 = h_2$



$P_2 = 1$ bar

$h_f = 47.9$ kJ/kg

$h_g = 1417.7$ kJ/kg

$h_2 = h_f + x(h_g - h_f)$

$313.2 = 47.90 + x(1417.7 - 47.8)$

$x = 0.194$

6 6 **16 / 20**

- **- 0 pts** Correct
- **- 3 pts** Wrong schematic of the system
- **- 3 pts** Wrong engineering model
- **- 5 pts** Wrong mass rate and energy rate balance
- **- 5 pts** Wrong quality
- ✓ **- 4 pts Wrong T-v diagram**
- **- 2 pts** Incompelet engineering model